Swati Aggrawal

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PROFESSIONAL SUMMARY

MSc Computer Science graduate from Sapienza University of Rome specializing in Artificial Intelligence and Machine Learning. Experienced in developing AI-powered applications, backend services, and machine learning solutions using Python, FastAPI, Docker, AWS, and cloud technologies. Hands-on experience with software development, REST APIs, computer vision, NLP, and Generative AI applications. Familiar with Java, frontend development, databases, and collaborative software engineering practices. Passionate about building scalable products, solving real-world problems, and learning new technologies across the software stack.

SKILLS

- **Programming :** Python, Java, SQL, C++, Bash
- **Software Engineering :** Object-Oriented Programming (OOP), Data Structures & Algorithms, Unit Testing, Agile/Scrum, Software Development Lifecycle (SDLC), Git
- **Backend & APIs :** FastAPI, RESTful APIs, Backend Development, API Integration, Microservices Fundamentals
- **Frontend :** React.js (Basic), HTML, CSS, JavaScript
- **Databases :** PostgreSQL, MySQL, Relational Databases
- **Cloud & DevOps :** Docker, AWS, Cloud Computing Fundamentals, GitHub, Linux
- **Development Tools:** Git, GitHub, GitHub Copilot, Claude, Cursor
- **AI/ML :** Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, LLMs, Transformer Models, OpenCV, Model Evaluation & Optimization

EDUCATION

Sapienza Università di Roma **May 2026**

Master of Science in Computer Science

- **Relevant Coursework:** Computer Vision, Deep Learning and Applied AI, Natural Language Processing, Machine Learning, Distributed Systems
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

PROJECTS

DocDelta AI-Powered Document Intelligence & Comparison Platform

- Developed and deployed cloud-native backend services for AI-powered document analysis.
- Collaborated with AI-powered development tools and modern software engineering practices to accelerate development and testing.
- Designed RESTful endpoints using FastAPI for automated document analysis and change tracking workflows.
- Developed backend workflows for document processing and semantic comparison.
- Validated API functionality and document-processing workflows through systematic testing and debugging.
- Integrated transformer-based ML models into application pipelines for automated inference.
- Implemented semantic matching to identify added, removed, and modified content across document versions.
- Containerized the application using Docker and deployed it on AWS ECS Fargate.
- Configured application monitoring and logging using AWS CloudWatch.
- **Technologies:** Python, FastAPI, Sentence Transformers, Docker, AWS ECS Fargate, CloudWatch, REST APIs

PosePal Real-Time Yoga Pose Classification System

- Developed a real-time computer vision application that classifies yoga poses from live webcam input.
- Built and integrated a CNN-based image classification model for multi-class pose recognition.

- Implemented image preprocessing, normalization, and data augmentation techniques to improve model performance.
- Integrated model inference into a lightweight web application for real-time predictions.
- Applied frame-level prediction smoothing to reduce prediction fluctuations during live inference.
- Developed and integrated a browser-based user interface for real-time webcam-based pose classification.
- **Technologies:** Python, TensorFlow, Keras, OpenCV, Scikit-Learn

Metal Defect Detection using Computer Vision

- Developed computer vision models for automated defect detection using CNN, SVM, and Random Forest.
- Built preprocessing, training, and evaluation pipelines.
- Compared deep learning and traditional ML approaches.
- Performed model evaluation using accuracy, confusion matrices, and performance analysis.
- **Technologies:** Python, TensorFlow, Keras, OpenCV, Scikit-Learn

RESEARCH EXPERIENCE

Continual Learning for Computer Vision (Master's Thesis) — Nov 2025

- Evaluated pretrained Vision Transformer architectures under Task-Incremental learning settings.
- Designed training, evaluation, and benchmarking pipelines using Python and PyTorch.
- Automated experiment tracking and performance analysis across continual learning scenarios.
- **Technologies:** Python, PyTorch, Vision Transformers, Computer Vision

Machine Learning for Air Pollution [IEEE Publication] — May 2023

- Published IEEE conference paper evaluating machine learning approaches for environmental prediction and monitoring.
- Evaluated supervised ML algorithms including Random Forest, SVM, Decision Trees, and Neural Networks across environmental datasets.
- Performed comparative analysis using model performance metrics to assess accuracy, robustness, and generalization.
- Presented scientific findings through structured research documentation and publication.